

Blind rigid-motion correction in head cone-beam CT with a flow-matching prior on the geometry bridge

Author One^a, Author Two^a, and Author Three^{a,b}

^aDepartment, Institution, Address, City, Country

^bDepartment, Institution, Address, City, Country

ABSTRACT

Patient motion during a cone-beam CT (CBCT) head scan corrupts the reconstruction with streaks and blurring, and correcting it is a *blind* inverse problem: neither the image nor the per-view rigid pose is known, and the measurement is bilinear in the two. We propose a blind motion-correction method built on a flow-matching prior trained on the **geometry bridge** — a path of FDK reconstructions of the measured scan along which the residual motion decays linearly to zero, running from the uncorrected reconstruction to the motion-free one. Every point on it is an actual reconstruction of the actual measured data, so the regression target is available in closed form as the exact derivative of the backprojector with respect to the projection geometry, and the prior is trained on precisely the manifold inference traverses. At test time a 50-step predictor–corrector loop alternates a prior step, a 6-DoF-per-view motion fit on the improved image, and a plug-and-play data-consistency step. On 30 held-out CQ500 patients with Akima-spline motion of 10 mm / 10° peak-to-peak it cuts the mean reprojection error from 6.12 mm to 0.28 mm and raises SSIM from 0.546 to 0.978, improving every patient and landing within 0.005 SSIM of the same reconstruction given the ground-truth motion.

Keywords: cone-beam CT, rigid motion correction, flow matching, generative priors, plug-and-play, inverse problems